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An enhanced mobile localization algorithm integrating multiple AUKF models for mixed indoor environments

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Abstract

The positioning technology based on ultra-wideband ranging has been widely applied in the field of indoor positioning due to its excellent localization capabilities. However, mixed line-of-sight (LOS) and non-LOS (NLOS) indoor environments severely constrain positioning accuracy. To address this issue, we propose an innovative algorithm based on the adaptive unscented Kalman filter (AUKF) and interactive multiple model (IMM), designed to significantly enhance positioning accuracy in mixed indoor environments by mitigating the impact of NLOS errors and inaccurate process noise. Firstly, recognizing the distinct characteristics of ranging errors in indoor environments, we develop LOS and NLOS ranging models separately. Based on these models, the unscented Kalman filters are constructed for LOS and NLOS environments to accurately simulate the mixed LOS/NLOS indoor environments. Secondly, determining the statistical characteristics of process noise is challenging, often leading to degraded filter performance. We address this issue by proposing an environment-based AUKF algorithm, which significantly enhances the robustness and accuracy of the positioning system. Finally, the environment-based AUKFs are integrated into the IMM framework to constrain NLOS errors and achieve precise positioning effectively. Simulations, open-source dataset validation and experimental results demonstrate that the proposed algorithm significantly enhances the accuracy and stability of mobile target positioning in mixed LOS/NLOS indoor environments.

Keywords: indoor positioning, NLOS, mixed indoor environments, inaccurate process noise, AUKF

1. Introduction

In the intelligence era, location-based services (LBS) have become closely intertwined with daily life and vital to national

defense, economic construction, and social life. As the backbone of LBS applications, location technology plays an indispensable role. With the maturation of mobile communication and positioning technology, there is a growing demand for precise indoor positioning [1]. While satellite positioning technology can offer satisfactory accuracy for outdoor positioning, its performance degrades significantly indoors due to diverse obstacles hindering signal propagation [2]. Traditional indoor positioning technologies encompass Bluetooth [3], Wi-Fi [4], radio frequency [5], and infrared positioning [6]. However, these methods are susceptible to interference from radio frequency signals, resulting in diminished positioning accuracy

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that may fail to meet the stringent requirements of indoor positioning. To overcome these challenges, ultra-wideband (UWB) technology has emerged as a focal point in the field of indoor positioning due to its inherent strengths, including strong anti-interference capability, superior signal penetration, and low power consumption [7].

In UWB indoor positioning technology, the distance between the mobile node (MN) and the beacon nodes (BNs) is first determined through UWB signals. Subsequently, the position of the MN is calculated based on these measured distances. In complex indoor environments, UWB signals propagate through both line-of-sight (LOS) and non-LOS (NLOS) environments. NLOS environments generally result in longer transmission times than LOS environments, leading to a significant positive deviation in the measured distance, known as the NLOS errors. The error characteristics in purely LOS or NLOS indoor environments are relatively simple. In the LOS environment, process noise is a major source of error, while in the NLOS environment, the main factor affecting accuracy is NLOS errors [8]. The actual indoor environment is often a mixture of LOS and NLOS environments, making signal transmission more complex and transitions between environments more frequent. Furthermore, the frequent and dynamic transitions between LOS and NLOS environments make it challenging to accurately characterize the process noise. Enhancing positioning accuracy in mixed LOS/NLOS environments requires addressing both NLOS errors and process noise. Therefore, improving positioning accuracy in mixed LOS/NLOS indoor environments remains a significant challenge.

To address it, we propose a novel interactive multiple model algorithm based on the adaptive unscented Kalman filter (IMM-AUKF) for UWB indoor positioning systems. Our contributions are summarized as follows:

- (1) Recognizing the distinct ranging errors characteristics in mixed LOS/NLOS indoor environments, we develop LOS and NLOS ranging models separately. The UKFs are constructed based on these models, effectively simulating the mixed LOS/NLOS indoor environments with improved accuracy;
- (2) Determining the statistical characteristics of process noise is challenging in mixed LOS/NLOS indoor environments, often leading to degraded filter performance. We address this issue by proposing an environment-based AUKF algorithm, significantly enhancing the robustness and accuracy of the positioning system;
- (3) To address the NLOS errors affecting positioning accuracy in mixed LOS/NLOS indoor environments, we propose an IMM-AUKF algorithm and apply it to the UWB location systems. This algorithm addresses the issue of poor positioning accuracy in existing UWB indoor positioning systems within mixed LOS/NLOS indoor environments, enabling more accurate localization of mobile targets.

2. Related works

Currently, many indoor positioning algorithms focus on improving positioning accuracy by mitigating NLOS errors, employing two primary approaches: identification and constraint. The identification methods aim to identify and discard ranging values affected by NLOS errors, employing only LOS ranging values for positioning. Guo *et al* [9] employ statistical methods such as residual checks to identify and remove NLOS-ranging values. Haigh *et al* [10] identify NLOS-ranging values utilizing outliers in the signal characteristics of direct and indirect paths. Liu *et al* [11] establish training features from the measured signals and employ fuzzy logic to identify and discard NLOS ranging values, relying solely on LOS ranging values for positioning. However, the complexity of indoor environments prevents the complete identification and exclusion of NLOS-ranging values. In strong NLOS environments, the identification methods may lead to positioning inaccuracy due to the scarcity of LOS ranging values. The constraint methods directly calculate all range values without pre-identifying and discarding NLOS range values, thereby mitigating positioning errors caused by such issues. Chen *et al* [12] improve positioning accuracy by constraining NLOS errors through an optimized weighted least squares algorithm. Zhao *et al* [13] propose a robust extended Kalman filter (EKF) and a higher-order UKF, with experimental results showing effective mitigation of NLOS errors. Furthermore, adaptive Kalman algorithms have been proposed to improve positioning performance and robustness in complex indoor environments [14].

Most of these methods primarily focus on single NLOS environments or relatively stable scenarios. However, the mixed LOS/NLOS indoor environments are harsh and changeable [15]. There are two main approaches to addressing mixed LOS/NLOS indoor environments: adding extra sensors and optimizing algorithms. In [16], an IMM algorithm combining UWB and inertial navigation is proposed to mitigate the impact of mixed LOS/NLOS indoor environments on positioning accuracy. In [17], an adaptive algorithm framework based on post-detection ranging values is used to develop an integrated positioning system that combines UWB, inertial measurement units, and odometry. These methods are costly, complex, and often constrained by device limitations. Moreover, most existing indoor positioning systems rely on UWB devices only, making algorithm optimization for improving performance in mixed LOS/NLOS indoor environments more practical. This paper also focuses on this core issue. IMM and KF-based optimization algorithms are common methods for addressing the impact of mixed LOS/NLOS indoor environments on positioning accuracy. Bian *et al* [18] propose a neural network-based multi-filter algorithm that uses KF and UKF for grouped calculations after identifying LOS and NLOS conditions. Yu *et al* [19] introduce a curve-fitting Kalman algorithm that significantly enhances positioning accuracy in complex indoor environments. Based

on the IMM, [20] combines semi-definite programming with an improved convex programming algorithm to reduce NLOS errors. Wang *et al* [21] optimize the IMM algorithm using fuzzy detection theory, with experiments demonstrating its effective suppression of NLOS errors.

The above algorithms help to improve indoor positioning accuracy. However, issues such as inaccurate identification, strong nonlinearity, and imprecise process noise statistics continue to impact positioning accuracy and remain inadequately addressed. To address these issues, we propose a novel IMM-AUKF algorithm. This algorithm directly processes all ranging values without pre-identifying and discarding NLOS ranging values, thereby mitigating positioning errors caused by inaccurate identification. Moreover, the AUKF models adapt to different environments within the algorithm, effectively handling strong nonlinearity and imprecise process noise in mixed LOS/NLOS indoor environments. In the IMM-AUKF algorithm, the AUKF models adapted for LOS and NLOS environments interact via a Markov chain, effectively suppressing complex and variable NLOS errors in mixed environments and ultimately enhancing positioning accuracy in mixed LOS/NLOS indoor environments.

3. System model

Due to the limitation of the MN in mixed LOS/NLOS indoor environments, we utilize the uniform acceleration model to describe the MN motion. Additionally, we analyze the errors in mixed environments and construct the corresponding ranging models.

3.1. MN motion model

$\hat{\mathbf{X}}(k) = [x(k), y(k), v_x(k), v_y(k)]^T$ is the MN state vector at time k in the localization scenario. $(x(k), y(k))$ denotes the MN position in a two-dimensional coordinate system. $(v_x(k), v_y(k))$ represents the velocity in the direction corresponding to the position. We use the alteration of the state vector to describe a MN motion process, as shown in equation (1):

$$\hat{\mathbf{X}}(k) = \mathbf{F}\hat{\mathbf{X}}(k-1) + \mathbf{G}\varpi(k-1) \quad (1)$$

where $\mathbf{F}$ and $\mathbf{G}$ represent the state transition and control matrix, respectively, as follows:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$\mathbf{G} = \begin{bmatrix} T^2/2 & 0 & T & 0 \\ 0 & T^2/2 & 0 & T \end{bmatrix}^T \quad (3)$$

where T is the sampling period. We assume the process noise $\varpi(k)$ is the Gaussian white noise with mean zero value and obeys the $N(0, \sigma_s^2)$. Its covariance matrix is $\mathbf{Q}(k)$ [22].

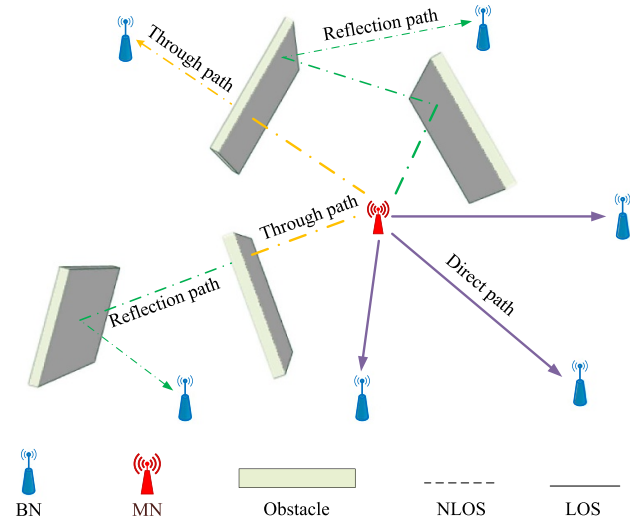


Figure 1. Signal transmission environment amid nodes.

3.2. Ranging model

3.2.1. Error analysis. Indoor positioning accuracy is typically influenced by factors such as NLOS errors, multipath effects, TOA/TDOA ranging methods, and BN geometric configuration. In mixed LOS/NLOS indoor environments, NLOS errors are the primary factor affecting positioning, and this paper addresses this issue. We propose an enhanced mobile indoor positioning algorithm based on UWB, leveraging its resistance to multipath effects. Separate ranging models are specifically developed for LOS and NLOS environments, tailored to changeable indoor settings. NLOS and systematic errors from TOA/TDOA ranging methods are incorporated as error compensations in these models to mitigate NLOS errors and process noise. This paper assumes that the geometric distribution of BN is consistent and therefore does not explore the impact of its distribution on positioning accuracy. Since this paper focuses on mitigating NLOS errors and process noise in mixed LOS/NLOS indoor environments, the discussion is centered on the relevant ranging models.

3.2.2. Ranging model construction. To better show the indoor LOS and NLOS signal transmission environment, we assume the MN moves within the BN monitoring area and that six BNs are distributed in the area to participate in the MN positioning process.

Figure 1 shows the signal transmission environment at some point during the MN moves in the area.

We assume that a total of N BNs detect the MN at time k and obtain the distance between the MN and BNs by the two-way ranging (TWR) method. When the signal transmission environment between the MN and m th BN is LOS, the ranging model is:

$$\mathbf{Z}_m(k) = \mathbf{d}_m(k) + \omega_m^{\text{los}}(k) \quad (4)$$

where $\mathbf{d}_m(k)$ is the true distance between the MN and m th BN at time k , $m \in [1, 2, \dots, N]$. $\omega_m^{\text{los}}(k)$ represents the ranging noise

in the LOS environment, primarily caused by clock synchronization errors, multipath effects, and environmental noise. It follows the Gaussian distribution $N(0, 1^2)$.

$$\mathbf{d}_m(k) = \sqrt{(x(k) - x_m(k))^2 + (y(k) - y_m(k))^2} \quad (5)$$

where $(x_m(k), y_m(k))$ represents the coordinates of the m th BN. When various obstacles block the signal transmission between a MN and the m th BN, the NLOS ranging model is:

$$\mathbf{Z}_m(k) = \mathbf{d}_m(k) + \omega_m^{\text{nlos}}(k) \quad (6)$$

where $\omega_m^{\text{nlos}}(k)$ indicates the ranging noise in the NLOS environment, including $\omega_m^{\text{los}}(k)$ and NLOS errors. Since $\omega_m^{\text{los}}(k) \sim N(0, \mathbf{1})$, $\omega_m^{\text{nlos}}(k)$ is set to follow a Gaussian distribution with a non-zero mean [22].

4. Proposed algorithm

We propose an IMM-AUKF algorithm to enhance positioning accuracy in mixed LOS/NLOS indoor environments, as shown in figure 2. Firstly, the ranging values are inputted into filters based on LOS and NLOS environments to filter processing. In this step, we utilize UKF to accommodate the nonlinearity of the indoor environment. For mixed LOS/NLOS indoor environments, we introduce the concept of smooth filtering to construct an adaptive matrix and optimize UKF into AUKF. It can reduce the influence of inaccurate statistical characteristics of process noise on filtering accuracy. Furthermore, we achieve an AUKF tailored for LOS and NLOS environments by adjusting it based on the distinct characteristics of ranging errors in different environmental settings.

Then, the IMM-AUKF algorithm is designed based on the AUKF adapted to the environment, which can effectively suppress NLOS errors. In this part, the AUKFs based on the LOS and NLOS environments are used as two parallel subfilters within the IMM framework. The transformation between different models is realized by the Markov transfer matrix. If the ranging value contains NLOS errors, the AUKF adapted to the NLOS environment will output relatively accurate filtering values, consequently increasing the corresponding model probability. Finally, the IMM-AUKF algorithm performs a weighted summation calculation of the filtering values from different models based on model probabilities, accurately locating moving targets in the mixed LOS/NLOS indoor environments.

4.1. AUKF algorithm

The MN system equation is:

$$\begin{cases} \hat{\mathbf{X}}(k) = \mathbf{F}\hat{\mathbf{X}}(k-1) + \mathbf{G}\varpi(k-1) \\ \mathbf{Z}(k) = \mathbf{d}_m(k) + \omega_m(k) \end{cases} \quad (7)$$

where $\omega_m(k)$ represents the ranging noise, as shown in equation (8),

$$\omega_m(k) = \begin{cases} \omega_m^{\text{los}}(k), & \text{if LOS} \\ \omega_m^{\text{nlos}}(k), & \text{if NLOS} \end{cases} \quad (8)$$

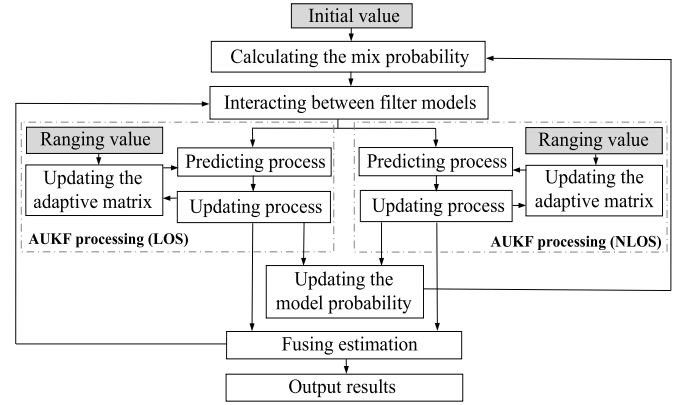


Figure 2. IMM-AUKF algorithm block diagram.

The AUKF algorithm consists of two parts: AUKF filtering and adaptive matrix updating.

4.1.1. AUKF filtering. A UKF can mitigate the errors arising from the nonlinearity of the indoor environment by using an unscented transformation of a series of sigma sampling points. Firstly, we specify the initial state value $\hat{\mathbf{X}}(k-1)$ and the covariance matrix $\mathbf{P}(k-1)$. Then, a set of sigma points is generated from a specified initial value to simulate the probability distribution of the nonlinear system.

$$\mathbf{P}_{\text{mid}}(k-1) = \sqrt{(n+\lambda)\mathbf{P}(k-1)} \quad (9)$$

$$\begin{cases} \xi^0(k-1) = \hat{\mathbf{X}}(k-1) \\ \xi^i(k-1) = \hat{\mathbf{X}}(k-1) + (\mathbf{P}_{\text{mid}}(k-1))_i, i = 1, 2, \dots, n \\ \xi^i(k-1) = \hat{\mathbf{X}}(k-1) - (\mathbf{P}_{\text{mid}}(k-1))_i, i = n+1, n+2, \dots, 2n \end{cases} \quad (10)$$

where $\mathbf{P}_{\text{mid}}(k-1)$ is an intermediate variable. And n indicates the dimensions of the positioning system, which is set to four. $\xi^i(k-1)$ represents the i th sigma point. $\lambda = \alpha^2(n+\kappa) - n$, where α can adjust the degree of dispersion of the sigma points. Generally, $10^{-4} \leq \alpha \leq 1$ and κ is usually set to zero.

Secondly, we calculate the one-step prediction $\xi^i(k)$. The state vector $\hat{\mathbf{X}}(k|k-1)$ and covariance matrix $\mathbf{P}(k|k-1)$ are obtained through weighted summation.

$$\xi^i(k) = \mathbf{F}\xi^i(k-1) \quad (11)$$

$$\hat{\mathbf{X}}(k|k-1) = \sum_{i=0}^{2n} W_i^m \xi^i(k) \quad (12)$$

$$\mathbf{E}(k|k-1) = (\xi^i(k) - \hat{\mathbf{X}}(k|k-1)) \quad (13)$$

$$\mathbf{P}(k|k-1) = \sum_{i=0}^{2n} W_i^c \mathbf{E}(k|k-1) \mathbf{E}^T(k|k-1) + \psi(k-1) \mathbf{G} \mathbf{Q}(k-1) \mathbf{G}^T \psi^T(k-1) \quad (14)$$

where W_i^m and W_i^c are the weighting factors for the mean and variance, respectively. $\psi(k)$ is the adaptive matrix,

$$W_i^m = \begin{cases} \frac{\lambda}{n+\lambda} & i = 0 \\ \frac{1}{2(n+\lambda)} & i = 1, 2, \dots, 2n \end{cases} \quad (15)$$

$$W_i^c = \begin{cases} \frac{\lambda}{n+\lambda} + 1 - \alpha^2 + \beta & i = 0 \\ \frac{1}{2(n+\lambda)} & i = 1, 2, \dots, 2n \end{cases} \quad (16)$$

Additionally, we substitute $\xi^i(k)$ into the observation equation $h(\bullet)$ yields the predicted value $\chi^i(k)$. Subsequently, we can compute the mean $\hat{Z}(k)$, corresponding variance $P_{yy}(k)$ and $P_{xy}(k)$ of the predicted values $\chi^i(k)$.

$$\chi^i(k) = h(\xi^i(k)) \quad (17)$$

$$\hat{Z}(k) = \sum_{i=0}^{2n} W_i^m \chi^i(k) \quad (18)$$

$$P_{yy}(k) = \sum_{i=0}^{2n} W_i^c (\chi^i(k) - \hat{Z}(k)) (\chi^i(k) - \hat{Z}(k))^T + R(k) \quad (19)$$

$$P_{xy}(k) = \sum_{i=0}^{2n} W_i^c (\xi^i(k) - \hat{X}(k|k-1)) (\chi^i(k) - \hat{Z}(k))^T \quad (20)$$

where $R(k)$ is the ranging noise covariance matrix.

Finally, we calculate the UKF gain $K(k)$. The innovation vector $\nu(k)$ is updated by the ranging value from the ranging models,

$$K(k) = P_{xy}(k) P_{yy}^{-1}(k) \quad (21)$$

$$\nu(k) = Z_m(k) - \hat{Z}(k) \quad (22)$$

$$\hat{X}(k) = \hat{X}(k|k-1) + K(k) \nu(k) \quad (23)$$

$$P(k) = P(k|k-1) - K(k) P_{yy}(k) K^T(k). \quad (24)$$

Additionally, the innovation vector $\nu(k)$ follows a Gaussian distribution with mean zero and covariance matrix $P_{yy}(k)$.

4.1.2. Adaptive matrix updating. When the innovation vector $\nu(k)$ becomes significant, $\nu(k)$ no longer obeys a Gaussian distribution. It could lead to $E(\nu(k) \nu(k)^T) > P_{yy}(k)$.

Assuming the initial values $\hat{X}(k-1)$, $P(k-1)$ and models of the system are accurate, we set $\Delta(k)$ to represent the covariance matrix of the errors due to inaccurate noise.

$$\Delta(k) = E(\nu(k) \nu(k)^T) - P_{yy}(k) \quad (25)$$

$V(k)$ is the covariance matrix of the innovation vector, as follows:

$$V(k) = \begin{cases} \nu(k) \nu(k)^T & k = 1 \\ \frac{\rho V(k-1) + \nu(k) \nu(k)^T}{1 + \rho} & k \geq 2 \end{cases} \quad (26)$$

where the forgetting factor ρ satisfies $0 < \rho < 1$.

These filtering results are incompletely accurate due to the inaccuracy of the initial value, the ranging model, and noise in the filtering process. Thus, $\Delta(k)$ may not be zero. We design a decision threshold η ($\eta > 0$) for precise correction.

The trace of $\Delta(k)$ is compared with η in real time to determine whether to adjust the process noise covariance matrix. When $tr(\Delta(k)) > \eta$, the deviation between $V(k)$ and $P_{yy}(k)$ becomes significant, and the covariance matrix of the process noise needs to be adjusted. Otherwise, adjustment is not necessary.

To make the algorithm stable, we design the adaptive matrix $\psi(k)$ to balance the connection between the covariance matrix $Q(k)$ and the innovation vector $\nu(k)$.

$$\psi(k) = \text{diag}[\varphi_1(k), \varphi_2(k), \varphi_i(k) \dots \varphi_n(k)] \quad (27)$$

where $\varphi_i(k)$ is the diagonal element of the $\psi(k)$.

$$\begin{cases} \varphi_i(k) = \sqrt{\lambda_1 \varphi_i^2(k-1) + \lambda_2 \frac{tr(\Delta(k))}{tr(Q(k))}}, & tr(\Delta(k)) > \eta \\ \varphi_i(k) = 1, & tr(\Delta(k)) \leq \eta \end{cases} \quad (28)$$

where, λ_1 and λ_2 are smoothing parameters, generally, $0 < \lambda_1 < 1$ and $\lambda_1 + \lambda_2 = 1$. As λ_1 increases, the AUKF filter becomes more stable, yet its capability to effectively process the ranging value diminishes. In contrast, as λ_2 increases, the AUKF demonstrates enhanced effectiveness in processing the ranging value. Consequently, the AUKF has better performance than UKF. Furthermore, even when the value of η is significant, the AUKF can still filter stably, albeit at the expense of increased computational overhead to derive parameter $\varphi_i(k)$.

4.2. IMM-AUKF algorithm

In mixed indoor environments, the signal transmission environment randomly transforms between LOS and NLOS. Hence, a single filtering model cannot process the ranging value effectively. We use an IMM to combine different filter models to match the mixed indoor environments.

Firstly, we calculate the mixing probability of different filtering models and normalize it.

$$u^{ij}(k-1|k-1) = \frac{\pi_{ij} u^i(k-1)}{\bar{c}_j} \quad (29)$$

$$\bar{c}_j = \sum_{i=1}^2 \pi_{ij} u^i(k-1) \quad (30)$$

where $u^{ij}(k-1|k-1)$ denotes the mixing probability of models i and j at time $(k-1)$. π is the Markov transition matrix. $u^i(k-1)$ represents the model probability of the model i . $\bar{c}_j$ is the normalization constant. Among $i, j \in [1, 2]$, 1 and 2 represent LOS and NLOS filtering models, respectively.

Secondly, the initial filtering values of each filtering model are obtained by state interaction.

$$\hat{X}^{0j}(k-1) = \sum_{i=1}^2 \hat{X}^i(k-1) u^{ij}(k-1|k-1) \quad (31)$$

$$E_2(k-1) = \hat{X}^j(k-1) - \hat{X}^{0j}(k-1) \quad (32)$$

$$P^{0j}(k-1) = \sum_{i=1}^2 u^{ij}(k-1|k-1)$$

$$\cdot [\mathbf{P}^i(k-1) + \mathbf{E}_2(k-1)\mathbf{E}_2(k-1)^T] \quad (33)$$

where $\hat{\mathbf{X}}^j(k-1)$ and $\mathbf{P}^j(k-1)$ are the estimated state value and covariance matrix of model j . $\hat{\mathbf{X}}^{0j}(k-1)$ and $\mathbf{P}^{0j}(k-1)$ are the mixing state estimates and the corresponding covariance matrix.

Thirdly, the AUKFs based on LOS and NLOS environments receive the initial filtering values and conduct the filter process simultaneously.

For two filters $\hat{\mathbf{X}}^{0j}(k-1)$, $\mathbf{P}^{0j}(k-1)$ and $\mathbf{Z}_m(k)$ are inputs to the AUKF (The AUKF process is illustrated in equations (9)–(28)). Then, the state estimates $\hat{\mathbf{X}}^j(k)$ of the LOS and NLOS models and the corresponding covariance matrix $\mathbf{P}^j(k)$ after the filtering update are obtained.

Fourthly, we update the model probabilities of filtering models based on LOS and NLOS environments.

$$u^j(k) = \frac{\Lambda^j(k)\mathbf{c}_j}{\mathbf{c}} \quad (34)$$

$$\mathbf{c} = \sum_{j=1}^2 \Lambda^j(k)\bar{\mathbf{c}}_j \quad (35)$$

where $\Lambda^j(k)$ is the likelihood functions of model j ,

$$\Lambda^j(k) = \frac{1}{\sqrt{2\pi \det(\mathbf{P}_{yy})}} \exp\left(-0.5 \left(\boldsymbol{\nu}^j(k)^T \mathbf{P}_{yy}^{-1} \boldsymbol{\nu}^j(k)\right)\right) \quad (36)$$

where $\det(\cdot)$ means to obtain the value of the determinant.

Finally, we integrate the state estimates output by filtering models based on different environments to obtain the final position estimate.

$$\hat{\mathbf{X}}(k) = \sum_{j=1}^2 u^j(k) \hat{\mathbf{X}}^j(k) \quad (37)$$

$$\mathbf{M}(k) = \left(\hat{\mathbf{X}}^j(k) - \hat{\mathbf{X}}(k) \right) \left(\hat{\mathbf{X}}^j(k) - \hat{\mathbf{X}}(k) \right)^T \quad (38)$$

$$\mathbf{P}(k) = \sum_{j=1}^2 u^j(k) (\mathbf{P}^j(k) + \mathbf{M}(k)) \quad (39)$$

$$\mathbf{L}(k) = \mathbf{B} \hat{\mathbf{X}}(k) \quad (40)$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad (41)$$

where $\mathbf{L}(k)$ represents the location information of the MN. $\mathbf{B}$ is the observation matrix.

5. Experimental validation and analysis

This section focuses on validating the superiority of the IMM-AUKF algorithm through simulations, open-source datasets and experiments. As noted in the introduction, IMM and KF-based algorithms are common and representative methods in mixed LOS/NLOS indoor environments positioning.

Therefore, IMM-UKF, IMM-EKF, AUKF, and UKF are selected as comparison algorithms to validate the performance of the IMM-AUKF. Additionally, we use root mean square error (RMSE) and cumulative distribution function (CDF) as evaluation metrics to effectively and intuitively assess the performance of the IMM-AUKF algorithm.

$$\text{RMSE} = \sqrt{\frac{1}{KM} \sum_{i=1}^M \sum_{k=1}^K \left((\hat{x}_i(k) - x_i(k))^2 + (\hat{y}_i(k) - y_i(k))^2 \right)} \quad (42)$$

where the sampling period $T = 1$ s; the sampling frequency is $K = 100$; and the number of Monte Carlo experiments is $M = 1000$. $(\hat{x}_i(k), \hat{y}_i(k))$ and $(x_i(k), y_i(k))$ represents the estimated and true positions of MN in the i th Monte Carlo experiment, respectively.

$$F(x) = P_{\text{CDF}}(\text{RMSE} < x) \quad (43)$$

where $F(x)$ gives the probability that the random variable RMSE takes on a value less than or equal to x .

5.1. Simulation

5.1.1. Simulation experiment setup. The simulation experiments are conducted using the MATLAB R2022a platform. The simulation environment is configured as follows: it is supposed that a MN moving within a $110 \text{ m} \times 110 \text{ m}$ area. Meanwhile, six BNs are randomly distributed within the region of MN movement. Figure 3 shows the trajectory of the MN and positions of these BNs. The initial state vector and covariance matrix of the MN are $\hat{\mathbf{X}}(0) = [0, 10, 1, 0.1]^T$ and $\mathbf{P}(0) = \mathbf{I}_4$, respectively. Furthermore, $\hat{\mathbf{X}}^j(0) = \hat{\mathbf{X}}(0)$ $\mathbf{P}^j(0) = \mathbf{P}(0)$. The initial probability $u^1(k-1)$ of the NLOS model in IMM is 0.5. As the sum of the NLOS and LOS model probabilities must be 1, the initial probability $u^2(k-1)$ of the LOS model in IMM is 0.5. The transition between the different filter models is accomplished via the Markov transition matrix of the IMM algorithm, with initial values set as follows:

$$\pi = \begin{bmatrix} 0.95 & 0.05 \\ 0.05 & 0.95 \end{bmatrix}. \quad (44)$$

Additionally, the NLOS probability P_N is designed to represent the intensity of the NLOS environment in mixed LOS/NLOS indoor environments. Adjusting P_N could effectively evaluate the algorithm's performance in different mixed environments.

$$P_N = \frac{\sum_{i=1}^{N_{\text{BN}}} T_{\text{NLOS}}^i}{T_{\text{sum}} \times N_{\text{BN}}} \quad (45)$$

where T_{NLOS}^i represents the total time the i th BN is in a NLOS environment during the positioning phase. T_{sum} is the time for positioning. N_{BN} represents the total number of BNs involved in positioning.

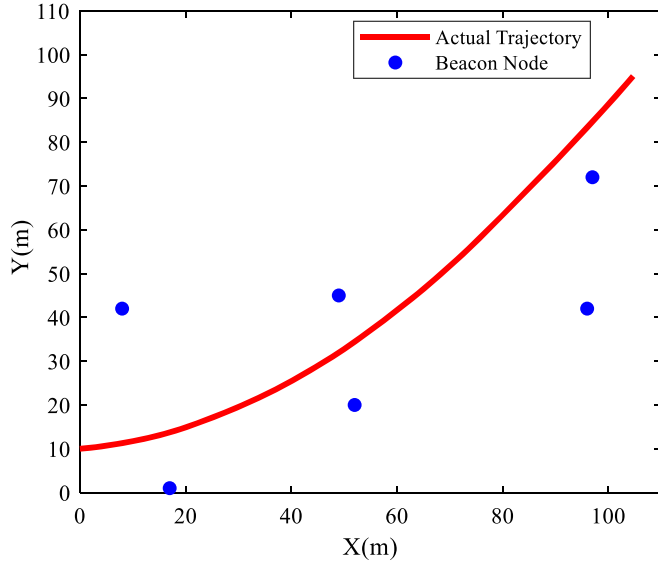


Figure 3. BN distribution and MN trajectory.

Table 1. Default values for experimental.

Parameter	Symbol	Default values
Number of BNs	N	6
NLOS probability	P_N	0.5
Ranging noise in LOS	$N(\mu_{\text{los}}, \sigma_{\text{los}}^2)$	$N(0, 1^2)$
Ranging noise in NLOS	$N(\mu_{\text{nlos}}, \sigma_{\text{nlos}}^2)$	$N(5, 6^2)$

Specifically, we assume that $\omega_m^{\text{los}}(k)$ and $\omega_m^{\text{nlos}}(k)$ follow Gaussian distribution $N(\mu_{\text{los}}, \sigma_{\text{los}}^2)$ and $N(\mu_{\text{nlos}}, \sigma_{\text{nlos}}^2)$, respectively. Table 1 provides the default values of the experimental parameters. In the scenario where $\omega_m^{\text{nlos}}(k)$ consistently follows a Gaussian distribution, we systematically vary one parameter at a time while keeping the other parameters constant. Three types of experiments are conducted in a mixed LOS/NLOS indoor environment, focusing on NLOS probability, the standard deviation of NLOS errors, and variations in the number of BNs. These experiments effectively validate the application and performance of the IMM-AUKF in mixed environments.

5.1.2. NLOS probabilities in mixed environments. To validate the performance of the IMM-AUKF algorithm in mixed LOS/NLOS indoor environments, we simulate various mixed conditions by adjusting the probability of the NLOS environment. $P_N = 0$ and $P_N = 1$ represents pure LOS and pure NLOS environments, respectively. P_N values between 0.1 and 0.9 represent mixed environments, where the NLOS environment ranges from weak to strong. Figure 4 shows that positioning errors for all algorithms increase as NLOS environment intensify. However, the IMM-AUKF consistently maintains the lowest error, demonstrating its superior performance. Even in pure LOS or NLOS environments, IMM-AUKF achieves relatively accurate positioning.

A comparison of the AUKF and UKF curves shows that the AUKF algorithm improves accuracy to some extent. However,

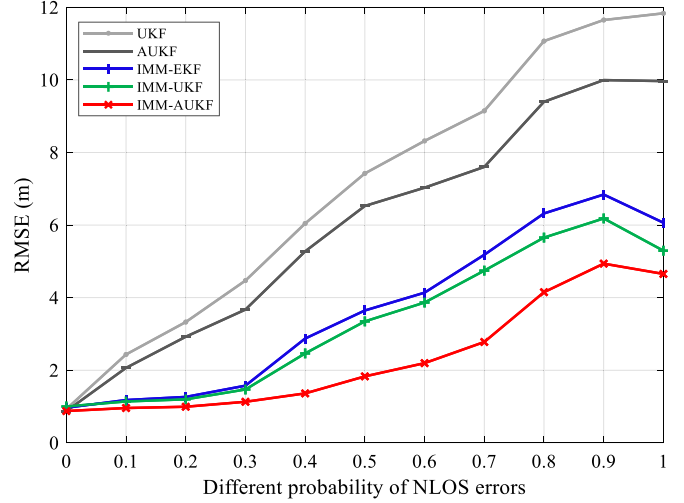


Figure 4. RMSE versus the different NLOS probability.

as the environment shifts from pure LOS to mixed, localization accuracy gradually decreases due to the inability to adapt to the mixed environments. Although AUKF mitigates the impact of inaccurate process noise, its struggle to suppress NLOS errors is limited. Therefore, this paper proposes the IMM-AUKF algorithm, combining IMM with AUKF to simultaneously reduce the effects of inaccurate process noise and NLOS errors in mixed LOS/NLOS indoor environments, thereby improving localization accuracy. Moreover, as the environment transitions from mixed to pure NLOS, errors across most algorithms tend to decrease, highlighting that mixed LOS/NLOS indoor environments are more complex than pure NLOS ones. This confirms the importance of the IMM-AUKF algorithm in improving positioning accuracy in complex mixed environments.

5.1.3. The standard deviation of NLOS errors in mixed environments. This section evaluates the performance of the IMM-AUKF algorithm under varying the standard deviation of NLOS errors in mixed environments. In the experiment, the standard deviation of NLOS errors gradually increases from 3 m to 12 m [23]. With the standard deviation increasing, the RMSE of the IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms shows an increasing trend, as illustrated in figure 5. The average positioning accuracy for IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms is 2.36 m, 4.12 m, 5.51 m, 9.12 m, and 10.06 m, respectively. IMM-AUKF demonstrates superior average positioning accuracy compared to IMM-UKF, IMM-EKF, AUKF, and UKF by 42.7%, 57.2%, 74.1%, and 76.5%.

5.1.4. The number of BNs in mixed environments. To assess the impact of BN redundancy information on positioning accuracy in mixed environments, we verify the IMM-AUKF algorithm by varying the number of BNs. Figure 6 illustrates

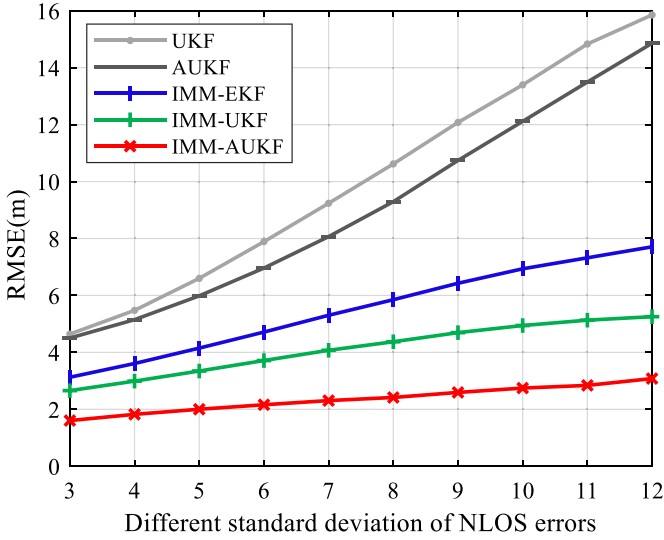


Figure 5. RMSE versus the different standard deviation.

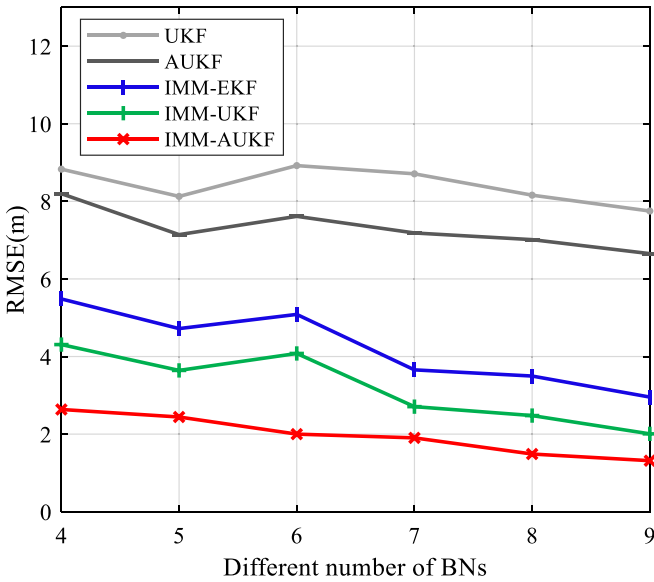


Figure 6. RMSE versus the different BNs.

the variation of RMSE for the IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms as the number of BNs increases.

The average positioning accuracies of the IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms are 1.96 m, 3.20 m, 4.23 m, 7.30 m, and 8.42 m, respectively. IMM-AUKF outperforms IMM-UKF, IMM-EKF, AUKF, and UKF in average positioning accuracy by 38.8%, 53.7%, 73.2%, and 76.7%, respectively.

5.1.5. CDF of positioning errors in mixed environments.

This section illustrates the approximate error distribution of each algorithm in mixed environments by plotting the CDF of errors under default settings, as shown in figure 7. The red curve represents the CDF for IMM-AUKF, which

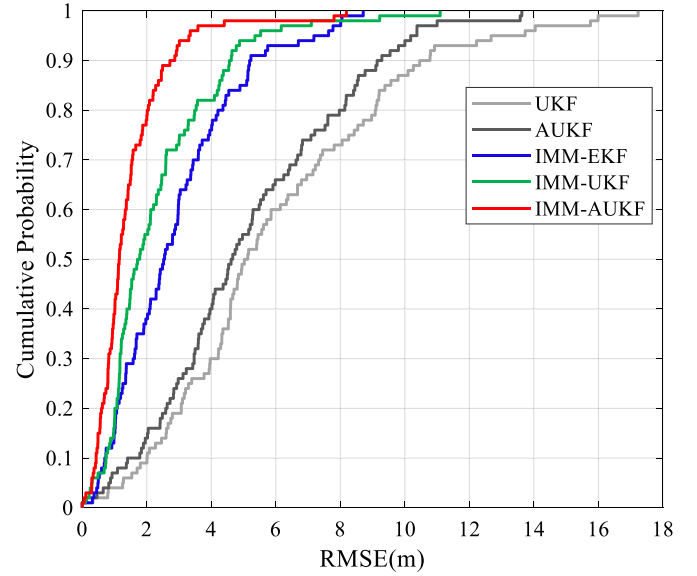


Figure 7. The CDF of the localization error.

consistently exhibits the most rapid convergence throughout the localization process. Moreover, the 90% RMSE of the IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms are below 2.72 m, 4.56 m, 5.19 m, 9.14 m, and 10.46 m, respectively. Based on both graphical and numerical analyses, it can be concluded that IMM-AUKF significantly outperforms the other algorithms in mixed environments.

5.2. Verification using open-source datasets

To further the validate the algorithm's performance, this section uses the open-source dataset from [24] in MATLAB R2022a to verify the proposed IMM-AUKF algorithm. For the experiments, the first six BNs are selected from Environment 0, which represents a residential-like indoor setting. Each BN collects 30 ranging measurements at each sampling point across all channels, with the average of the Channel 1 measurements used as the experimental ranging data. The dataset version used is 0.1.

Figures 8 and 9 show the RMSE and CDF results for various algorithms on the open-source dataset. The results are consistent with the simulations, demonstrating that the IMM-AUKF algorithm consistently outperforms the others. Figure 8 shows that the AUKF and UKF algorithms begin to diverge after the 60th sampling point due to environmental changes. As show in figure 9, for 70% of cases, the RMSE of the IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms are within 0.39 m, 0.69 m, 0.99 m, 2.88 m, and 3.21 m, respectively. These open-source experiments provide a more objective and comprehensive demonstration of the IMM-AUKF algorithm's superior performance in mixed LOS/NLOS indoor environments.

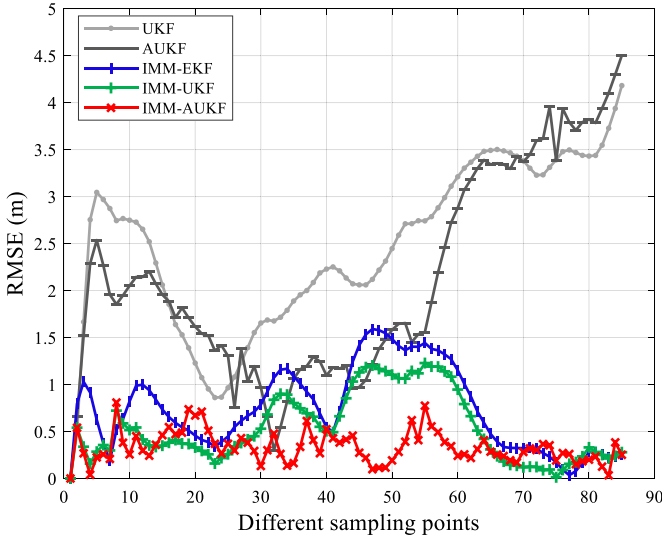


Figure 8. RMSE results under the open-source dataset.

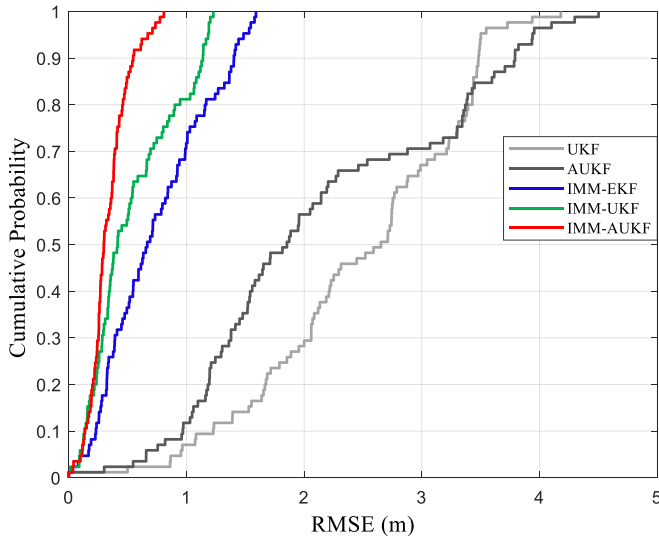


Figure 9. CDF results under the open-source dataset.

5.3. Experiment

Simulations and open-source datasets have verified that the IMM-AUKF algorithm accurately locates moving targets in mixed LOS/NLOS indoor environments. To further validate the performance and reliability of the IMM-AUKF algorithm, tests are conducted in real indoor scenarios.

5.3.1. Experimental setup. In the experiment, the MN and BN are implemented using DWM1000 modules. As shown in figure 10, the DWM1000 is a UWB transceiver module based on Decawave's DWM1000 chip. It integrates the antenna, RF circuits, power management, and clock circuits and supports the TWR ranging method. The module offers a range of approximately 100 m and a data transfer rate of up to 6.8 Mbps. Ranging information obtained through communication between the MN and BN is transmitted to a host computer

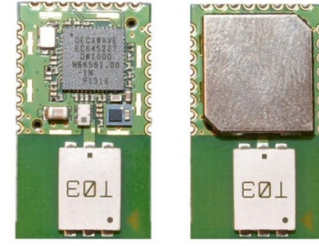


Figure 10. DWM1000 module.

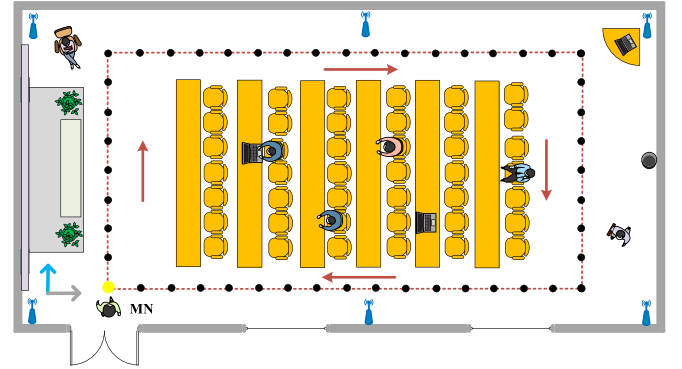


Figure 11. Experiment scene.

via the BN closest to the MN for processing. The host computer is a laptop running Windows 10, with an Intel i5-11300H CPU. DB Browser for SQLite (DB4S) is used to store ranging information, and the algorithm is executed on the MATLAB R2022a platform.

To conduct positioning experiments for the MN in a typical mixed LOS/NLOS indoor environment, considering factors such as ample space and ease of positioning equipment, we choose the conference room of the Science Hall in Dalian Maritime University as the experimental site. The conference room is a rectangular space measuring 10 m in length and 7 m in width. Six BNs are deployed to delineate the area for localization, ensuring sufficient movement range for the test subjects. The experiments are conducted with the southwest corner of the conference room as the origin, with the north direction as the positive y -axis direction and the east direction as the positive x -axis direction, as indicated by the blue and gray arrows in figure 11, respectively, establishing a two-dimensional Cartesian coordinate system. Figure 11 shows an experimental scenario after deploying the device. The blue markers represent the positions of BNs, while MN denotes pedestrians wearing the MN device. The yellow point represents the initial sampling point. MN begins its movement from this starting point along the red dashed line, taking distance measurements at the black sampling points during the journey, and finally returns to the yellow point.

In this experiment, the NLOS environment in a mixed LOS/NLOS indoor setting is constructed by placing fixed obstacles in front of the BNs, as shown in the top left and top right corners of figure 11. To mitigate interference from

ground reflections and refractions during UWB signal transmission, all BNs and the MN are positioned at a certain distance above the ground. In addition, the height of the BN deployed in the test is consistent with the MN worn by the tester. So, it is unnecessary to consider the change in the target's height to be positioned during the positioning process.

5.3.2. Data collection and processing. To experiment with the proposed IMM-AUKF algorithm, the equipment is first deployed as outlined in section 5.3.1, followed by data collection and processing. This process can be divided into three main parts:

- (1) **System Calibration:** first, remove all obstacles and fix the MN in a stationary position. Run the system to obtain the ranging values between the MN and all BNs. Then, manually measure the distances between the MN and each BN. Set the measurement indicator to 0.5 m based on the UWB device performance. If the difference between the system's ranging values and the manual measurements exceeds 0.5 m, the system is considered misaligned and requires readjustment. If the difference is within this range, the system is successfully calibrated.
- (2) **Data acquisition:** to ensure the validity of the collected data and avoid the influence of uncertain factors such as equipment failure on the positioning test, we set the MN to conduct ranging measurements with the six BNs every 0.5 m of movement. A total of 49 sampling points is distributed along a rectangular trajectory measuring 9 m in length and 6 m in width. Each BN performs 30 measurements per sampling, with the average value being utilized as the final ranging result. Additionally, the true positions of the deployed BNs are manually measured and used as reference data.
- (3) **Data Processing:** the collected ranging values and reference data are stored in the DB4S database on the host computer. MATLAB is used to access the data, run the algorithm, and estimate the MN's position.

Figure 12 shows the overall operational workflow of the UWB indoor positioning system.

The specific positioning process is as follows:

- (1) MN initiates distance measurement requests to all communicable BNs within the designated area via UWB communication, acquiring distances between MN and all communicable BNs;
- (2) Manually measure the distances between the MN and all BNs at the current time and calibrate the system by comparing them with the ranging values from step (1);
- (3) After calibration, the new ranging values are sorted in ascending order. The first BN in the sorted list, representing the nearest to the MN, is selected;
- (4) The MN sends all obtained ranging values to the nearest BN, which then forwards them to the host computer;
- (5) The host computer stores the system's ranging values and manually measured true values in the DB4S database.

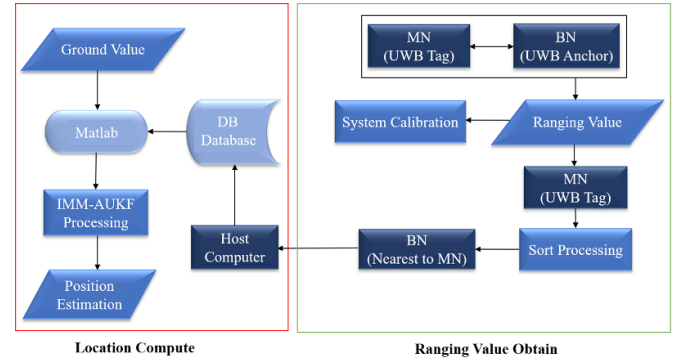


Figure 12. The operational flow of the UWB indoor positioning system.

MATLAB then processes this data using the positioning algorithm to determine the MN's position.

5.3.3. Experimental results and analysis. Based on the UWB indoor positioning system, we will conduct MN positioning experiments in the mixed LOS/NLOS indoor environments to further validate the performance of the IMM-AUKF algorithm. The initial state of MN is $\hat{\mathbf{X}}(0) = [1, 1, 0, 1]^T$. A mixed LOS/NLOS indoor environments is created by placing fixed obstacles in front of two BNs. All other parameters remain consistent with those in the simulation experiments. Figure 13 shows the RMSE of each algorithm during the localization process. The IMM-AUKF algorithm consistently outperforms the IMM-UKF, IMM-EKF, AUKF, and UKF algorithms in most cases. The average positioning accuracies for IMM-AUKF, IMM-UKF, IMM-EKF, AUKF, and UKF algorithms is 0.28 m, 0.36 m, 0.43 m, 0.93 m, and 0.98 m, respectively. Based on the average positioning accuracy, it can be observed that IMM-AUKF improves by 22.2%, 34.9%, 69.9%, and 71.4% compared to the IMM-UKF, IMM-EKF, AUKF, and UKF, respectively. Notably, the RMSE of IMM-AUKF at the 15th sampling point is 0.03 m. Referring to figure 11, we attribute this to two factors: most of the distance measurements at this point are LOS, and IMM-AUKF performs optimally here, resulting in minimal error.

To clearly show the distribution of positioning errors for each algorithm in mixed LOS/NLOS indoor environments, figure 14 shows the CDF of RMSE for each algorithm. IMM-AUKF exhibits the fastest convergence rate, with 90% of RMSE values distributed within 0.46 m. The IMM-UKF, IMM-EKF, AUKF, and UKF algorithms reach the 90% within 0.69 m, 0.86 m, 1.30 m, and 1.51 m, respectively.

Considering the impact of factors such as redundant information and the intensity of the NLOS environment in mixed environments, we conduct experiments with 4 BNs and 5 BNs. In these experiments, fixed obstacles are placed in front of two BNs to create the mixed environments, with all other parameters kept constant. The experimental results for different BN quantities are shown in Table 2. As indicated, the proposed IMM-AUKF algorithm delivers the

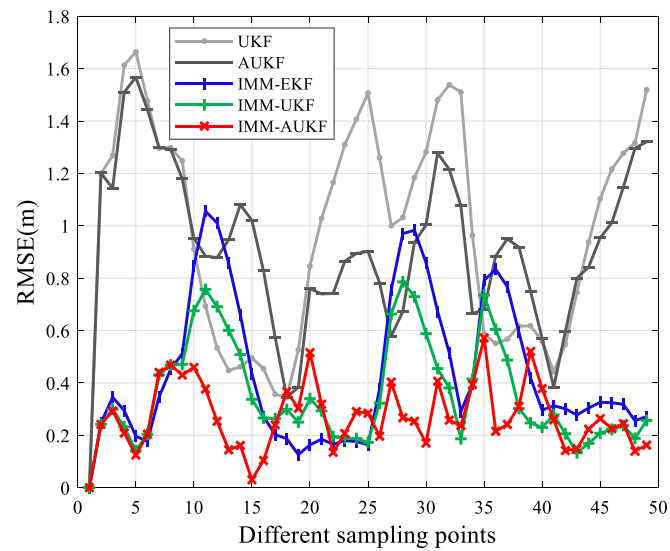


Figure 13. RMSE of sample points.

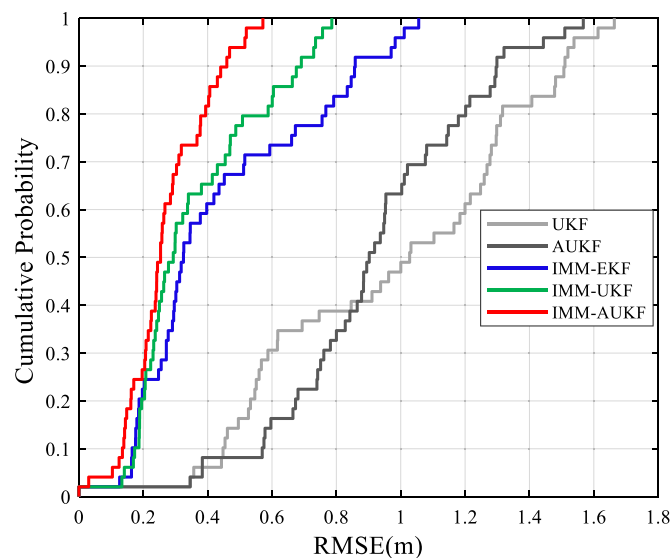


Figure 14. CDF of RMSE.

Table 2. Experimental results under different BN quantities.

BN quantity	Algorithm	Average RMSE (m)	RMSE at CDF = 90%
4	UKF	1.09	2.05
	AUKF	0.97	1.73
	IMM-EKF	0.57	1.17
	IMM-UKF	0.42	0.77
	IMM-AUKF	0.31	0.59
5	UKF	1.06	1.85
	AUKF	0.94	1.78
	IMM-EKF	0.47	0.97
	IMM-UKF	0.37	0.74
	IMM-AUKF	0.30	0.56

best performance, with each algorithm's results being consistent with the conclusions drawn from the previous experiments. Furthermore, increasing the number of BNs from 4 to 5 provides more redundant distance measurement information, while the intensity of the NLOS environment in the mixed LOS/NLOS indoor environments slightly decreases. As a result, all algorithms achieve lower RMSE values with 5 BNs compared to 4 BNs. These results strongly validate the effectiveness and robustness of the IMM-AUKF algorithm in mixed LOS/NLOS indoor environments, considering different redundant information and NLOS environment intensity.

6. Conclusions

To achieve better positioning of a mobile target in mixed LOS/NLOS indoor environments, considering the advantages of UWB technology such as anti-interference, strong penetration, and low power consumption, we research indoor positioning algorithms based on UWB positioning systems. Firstly, based on the characteristics of ranging errors, we establish UKFs based on LOS and NLOS environments to simulate the mixed LOS/NLOS indoor environments. However, the statistical characteristics of process noise cannot be accurately determined in complex indoor environments, leading to poor filtering performance. Drawing inspiration from smoothing filtering, we propose an AUKF based on innovation vectors, which adjusts the covariance matrix of process noise through an adaptive matrix to ensure stable and effective filtering. In indoor positioning, NLOS errors are the main factor affecting positioning accuracy. To address it, we innovatively propose an IMM-AUKF algorithm by combining the IMM with AUKFs. To validate the performance of the proposed IMM-AUKF algorithm, we design experiments in both simulation and actual indoor. Simulation and experimental results demonstrate that the IMM-AUKF algorithm can effectively constrain NLOS errors and improve the positioning accuracy of the mobile targets in mixed LOS/NLOS indoor environments.

In future research, we will delve deeper into the positioning of targets with complex mobility in special scenarios. Additionally, different obstacles may lead to variations in NLOS errors, thus affecting positioning accuracy. Future studies could involve extensive testing of various obstacles to investigate their impact on positioning.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Author contributions

Conceptualization, Y J and H G; methodology, H G; software, P Z; validation, Y J, H G and P Z; formal analysis, Y J and H G; investigation, H G; resources, Y J; data curation, P Z; writing—original draft preparation, H G; writing-review and editing, H G and Y J; visualization, Q H; supervision, Q H; project administration, Y J All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare no conflict of interest.

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